CS 391
Web Application Development

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Week - 2

Outline

1. HTML Graphics
2. HTML Media
3. Html Layout Elements

Html Graphics

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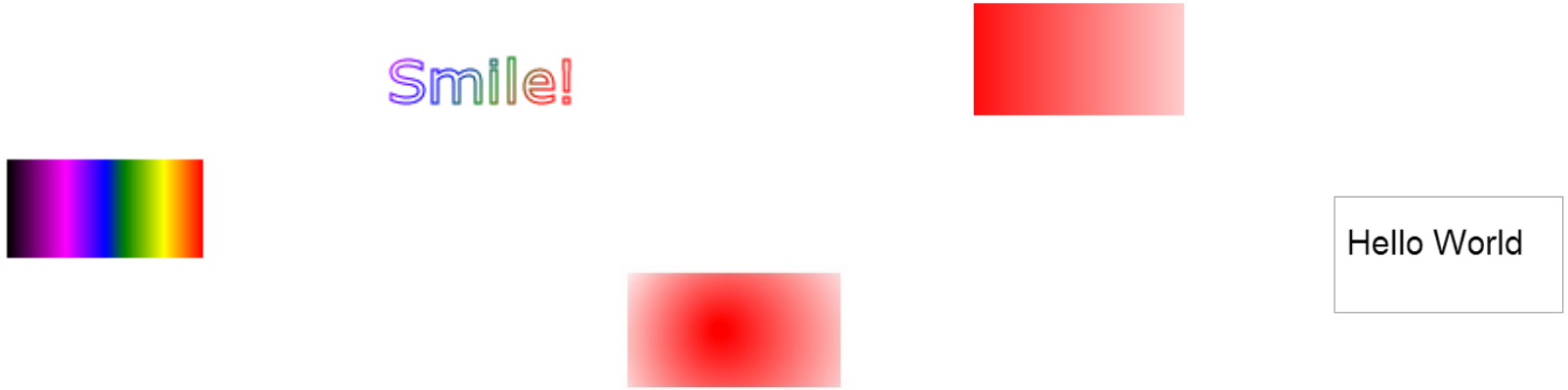
What is Canvas?

- <canvas> is an HTML element which can be used to draw graphics via scripting (usually JavaScript).
  - Dynamic, Scriptable rendering of 2D images
  - Resolution-dependent bitmap



https://www.w3schools.com/graphics/canvas_intro.asp

How to draw rectangles

- Three functions
  - fillRect(x, y, width, height) - creates a filled rectangle.
  - strokeRect(x, y, width, height) - creates a rectangular outline.
  - clearRect(x, y, width, height) - makes the rectangular transparent.

A Simple Canvas Example

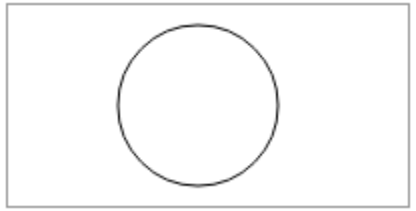
```
<script>
      var c=document.getElementById("myCanvas");
      var ctx=c.getContext("2d");
      ctx.fillStyle="#FF0000";
      ctx.fillRect(0,0,150,50);
</script>
```

drawing space:

```
<canvas id="myCanvas" width="200" height="100"
    style="border:1px solid #c3c3c3;">
</canvas>
```

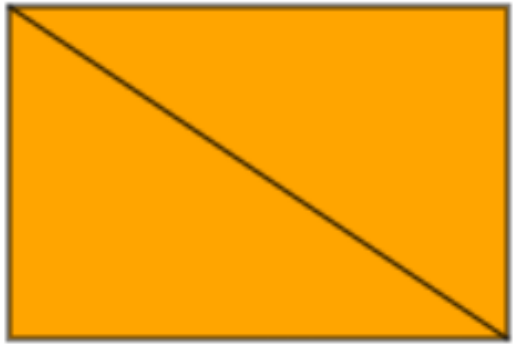
Canvas Example: Drawing a Circle

```
<script>
        var c=document.getElementById("myCanvas");
        var ctx=c.getContext("2d");
        ctx.beginPath();
        ctx.arc(95,50,40,0,2*Math.PI);
        ctx.stroke();
</script>
```



Canvas Example: Drawing a Line

```
function draw() {
        var canvas = document.getElementById('canvas');
        if (canvas.getContext) {
            const canvas = document.getElementById("myCanvas");
            const ctx = canvas.getContext("2d");
            ctx.fillStyle = 'orange';
            ctx.fillRect(30, 30, 150, 100);
            ctx.strokeRect(30, 30, 150, 100);
            ctx.moveTo(30, 30);
            ctx.lineTo(180, 130);
            ctx.stroke();            }
    }
```



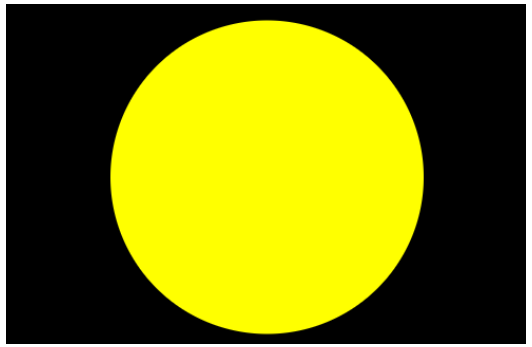
SVG

- SVG: **S**calable **V**ector **G**raphics
  - XML-based language for describing vector images
  - for marking up graphics, not content.
- SVG images are called **scalable** because their size can be increased or decreased without losing image quality.
  - because vector graphics consist of paths rather than pixels

https://www.w3schools.com/graphics/svg_intro.asp

SVG Example

```
<svg version="1.1" baseProfile="full" width="300" height="200"
                      xmlns="http://www.w3.org/2000/svg">
    <rect width="100%" height="100%" fill="black" />
    <circle cx="150" cy="100" r="90" fill= "yellow" />
</svg>
```



SVG vs. Canvas: Main Differences

Canvas	SVG
Resolution dependent	Resolution independent
No support for event handlers	Support for event handlers
Poor text rendering capabilities	Good text rendering capabilities
You can save the resulting image as .png or .jpg	Slow rendering if complex (anything that uses the DOM a lot will be slow)
Well-suited for graphic-intensive games	Not suited for game applications

For More Info

- Canvas
  - https://www.bitdegree.org/learn/html5-canvas-tutorial
  - https://www.w3schools.com/html/html5_canvas.asp

- SVG:
  - https://www.bitdegree.org/learn/html5-svg
  - https://www.w3schools.com/html/html5_svg.asp

Html Media

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Media Elements

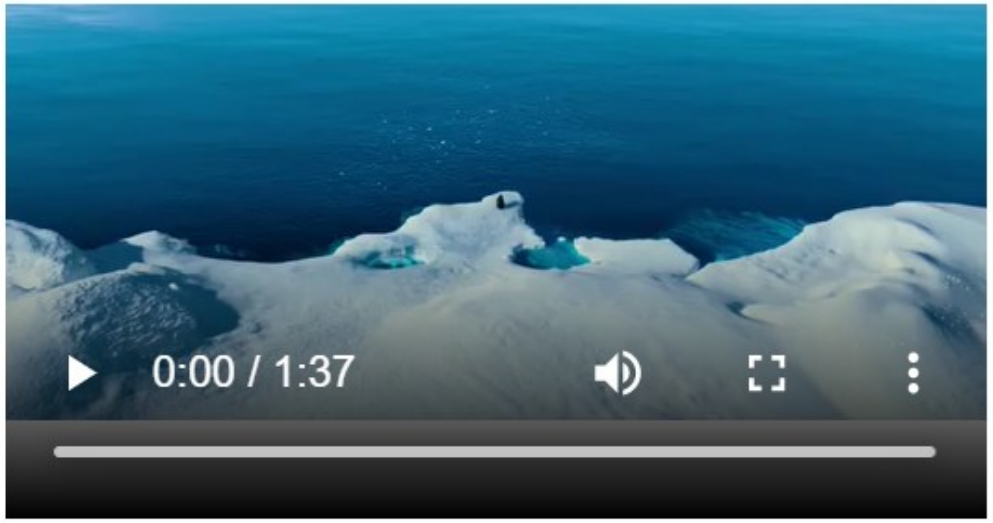
- Until HTML5
  - there has not been a standard for showing a video/movie.
  - through a plug-in (like flash).
  - different browsers may have different plug-ins.
- HTML5 defines a new elements
  - <video>
  - <audio>

Video

```
<video controls  width="320" height="200" autoplay  muted>
    <source src="happyfit2.mp4" type="video/mp4">

</video>
```

Attributes: autoplay, controls, loop, muted, height, width, src



Audio

```
<audio autoplay controls >
      <source type="audio/mp3" src="sound.mp3">
</audio>
```

Attributes: autoplay, controls, loop, src

Html Layout Elements

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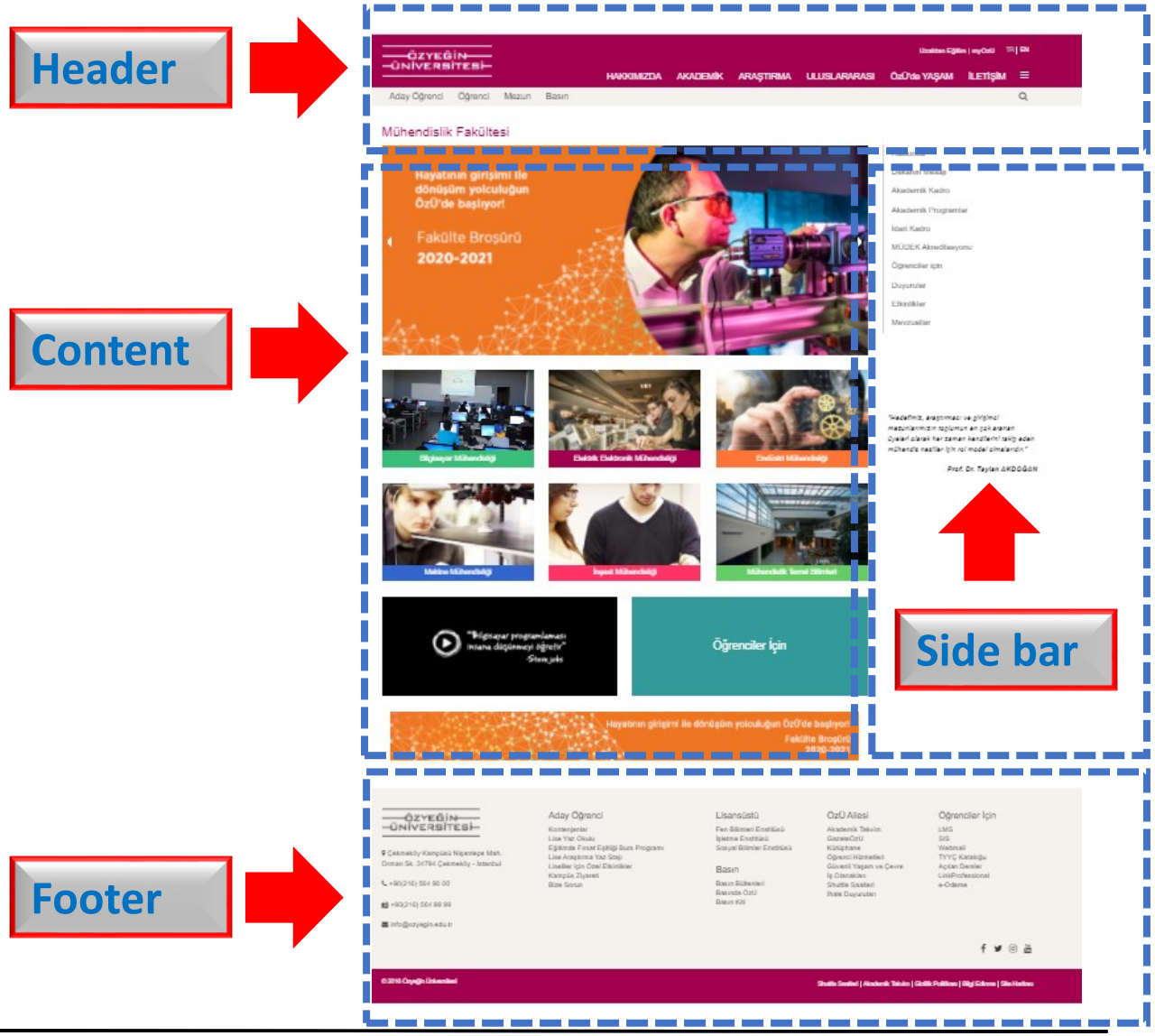
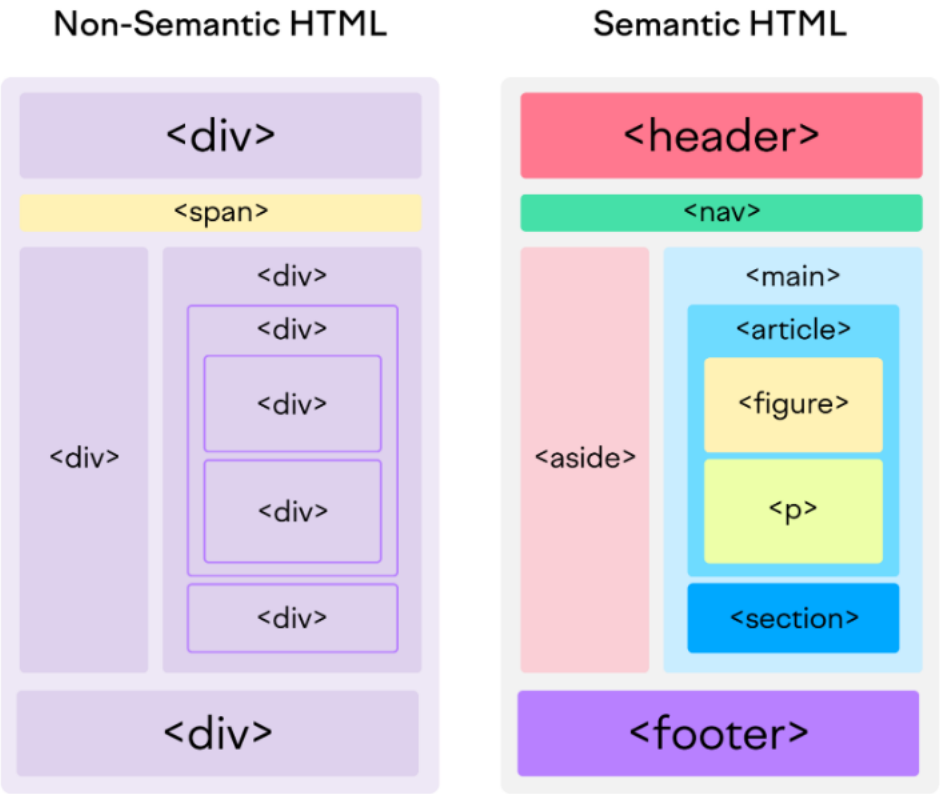
Layouts

- Design your webpage's layout very carefully.
- DO NOT use tables for layouts.
  - Tables are designed to display data in tabular form
- Use new semantic elements
  - Using **DIV** elements is old style ...



What are the Semantic Elements?

- **semantic** elements:
  - Elements with meaning
  - Clearly defines its content
  - <form>, <table>, <img>
- **non-semantic** elements:
  - Tells nothing about its content.
  - and



*A **div** element is used for block-level organization and styling of page elements, whereas a **span** element is used for inline organization and styling*

img: https://www.semrush.com/blog/semantic-html5-guide/

Semantic & non-semantic elements example

- This block of code uses ***semantic elements***:

```
<header></header>
<section>
    <article>
        <figure>
            <img>
            <figcaption></figcaption>
        </figure>
    </article>
</section>
<footer></footer>
```

- This second block of code uses ***non-semantic*** elements

```
<div id="header"></div>
<div class="section">
    <div class="article">
        <div class="figure">
            <img>
            <div class="figcaption"></div>
        </div>
    </div>
</div>
<div id="footer"></div>
```

Semantic Elements: <header> and <footer>

header for a document or section

```
<header>
   <h1>Ozyegin University</h1>
</header>
```

footer for a document or section

```
<footer>
   <p>Çekmeköy Campus, Nişantepe</p>
</footer>
```

Semantic Elements: <nav>

- Section of navigational links

```
<nav>
      <a href="/overview/">Overview</a>
      <a href="/faculty/">Faculty Information</a>
      <a href="/programs/">Academic Programs</a>
      <a href="/announcements/">Announcements</a>
</nav>
```

Semantic Elements: <section> and <article>

- Both these elements are used for sectioning a content, they can definitely be used interchangeably.
- HTML4 offered only one type of container element, which is ** .** While this is still used in HTML5, HTML5 provided us with **<section>** and **<article>** in a way to replace ** **

Semantic Elements: <section> and <article>

- The <section> tag defines sections in a document, such as chapters.

```
<section>
    <h1>WWW</h1>
    <p>The World Wide Web</p>
</section>
```

Semantic Elements: <section> and <article>

- <article> represents a component of a page that consists of a self-contained content
  - Forum post
  - Blog post
  - News story
  - Comment

```
<article>
      <h2>How to Bake a Cake</h2>
      <p>This is a step-by-step guide on how to bake a cake.</p>
</article>
```

More Info

- https://www.freecodecamp.org/news/semantic-html5-elements/
- https://www.semrush.com/blog/semantic-html5-guide/

Any Question?

Thank you for your attention

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